

TITLE: Wireshark Capture and Analysis

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SUMMARY

I captured the network traffic of my Kali system with concentration on its communication with an OWASPBWA VM. The tool I utilized in achieving this exercise is Wireshark.

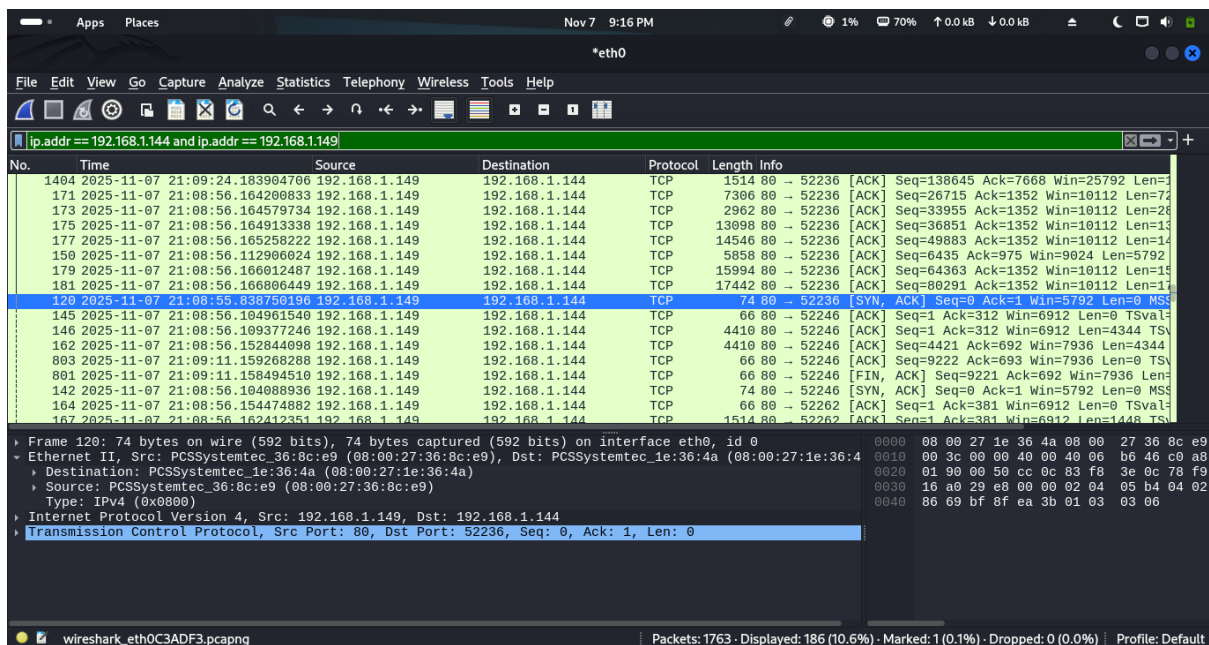
Wireshark captured over a thousand packets in under five minutes and during this time, I visited the damn vulnerable web application (dvwa) section of the OWASPBWA website via port 80.

ANALYSIS

Packets: 1763 · Displayed: 1763 (100.0%) · Marked: 1 (0.1%) · Dropped: 0 (0.0%)

The total captured packets are seen above to be **1763** and no packet was dropped.

After scrolling through the packets for observations, I filtered using my host's IP address - **192.168.1.144** and the OWASPBWA's - **192.168.1.149**. The total packets transmitted between the two endpoints are 186 and were transmitted through **TCP** and **HTTP** protocols.



After connection has been established between the two computers as can be seen from the TCP handshakes, the homepage of the OWASPBWA was reached via HTTP.

190	2025-11-07	21:09:30	100019004	192.168.1.149	192.168.1.144	TCP	74 89 → 32272 [STN, ACK] Seq=0 ACK=1 Win=3792 Len=0 MSS=
120	2025-11-07	21:09:35	339362277	192.168.1.144	192.168.1.149	HTTP	419 GET / HTTP/1.1
144	2025-11-07	21:08:56	104306138	192.168.1.144	192.168.1.149	HTTP	377 GET /animatedcollapse.js HTTP/1.1
187	2025-11-07	21:08:56	179442122	192.168.1.144	192.168.1.149	HTTP	325 GET /animatedcollapse.js HTTP/1.1
400	2025-11-07	21:09:01	967473389	192.168.1.144	192.168.1.149	HTTP	455 GET /dvwa HTTP/1.1
403	2025-11-07	21:09:02	018747593	192.168.1.144	192.168.1.149	HTTP	456 GET /dvwa/ HTTP/1.1
424	2025-11-07	21:09:02	319321363	192.168.1.144	192.168.1.149	HTTP	540 GET /dvwa/dvwa/css/login.css HTTP/1.1

Accept-Language: en-US,en;q=0.5\r\n	0000 00 00 27 36 8c e9 00 00 27 1e 36 4d
Accept-Encoding: gzip, deflate\r\n	0010 01 95 fe 32 40 00 00 06 b6 ba c9 a1
Connection: keep-alive\r\n	0020 01 95 cc 0c 00 50 78 f9 4b 59 83 f1
Upgrade-Insecure-Requests: 1\r\n	0030 01 f6 85 fd 00 00 01 01 08 0a bf 81
DNT: 1\r\n	0040 86 69 47 45 54 20 2f 20 48 54 54 50
Sec-GPC: 1\r\n	0050 0d 0a 48 6f 73 74 3a 20 31 39 32 2e
Priority: u=0, i\r\n	0060 31 2e 31 34 39 0d 0a 55 73 65 72 2c
\r\n	0070 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 3e
[Full request URI: http://192.168.1.149/]	0080 58 31 31 3b 20 4c 69 6e 75 78 20 7f
[HTTP request 1/10]	0090 34 3b 20 72 76 3a 31 32 38 2e 30 2f
[Response in frame: 130]	00a0 6b 6f 2f 32 30 31 30 30 31 30 31 2c
[Next request in frame: 137]	00b0 66 6f 78 2f 31 32 38 2e 30 0d 0a 4c

No.	Time	Source	Destination	Protocol	Length	Info
403	2025-11-07 21:09:02	018747593 192.168.1.144	192.168.1.149	HTTP	456	GET /dvwa/ HTTP/1.1
424	2025-11-07 21:09:02	319321363 192.168.1.144	192.168.1.149	HTTP	540	GET /dvwa/dvwa/css/login.css HTTP/1.1
1150	2025-11-07 21:09:16	813060459 192.168.1.144	192.168.1.149	HTTP	539	GET /dvwa/dvwa/css/main.css HTTP/1.1
427	2025-11-07 21:09:02	328517848 192.168.1.144	192.168.1.149	HTTP	602	GET /dvwa/dvwa/images/login_logo.png HTTP/1.1
1160	2025-11-07 21:09:16	834426928 192.168.1.144	192.168.1.149	HTTP	596	GET /dvwa/dvwa/images/logo.png HTTP/1.1
1156	2025-11-07 21:09:16	817813363 192.168.1.144	192.168.1.149	HTTP	526	GET /dvwa/dvwa/js/dvwaPage.js HTTP/1.1
1167	2025-11-07 21:09:16	800366646 192.168.1.144	192.168.1.149	HTTP	460	GET /dvwa/dvwa/js/dvwaPage.js HTTP/1.1
1177	2025-11-07 21:09:17	014438145 192.168.1.144	192.168.1.149	HTTP	584	GET /dvwa/favicon.ico HTTP/1.1
1134	2025-11-07 21:09:16	620475257 192.168.1.144	192.168.1.149	HTTP	609	GET /dvwa/index.php HTTP/1.1
406	2025-11-07 21:09:02	069513353 192.168.1.144	192.168.1.149	HTTP	525	GET /dvwa/login.php HTTP/1.1
1403	2025-11-07 21:09:24	177459150 192.168.1.144	192.168.1.149	HTTP	622	GET /dvwa/vulnerabilities/xss_r/ HTTP/1.1
1649	2025-11-07 21:09:34	623965209 192.168.1.144	192.168.1.149	HTTP	680	GET /dvwa/vulnerabilities/xss_r/?name=%3Cscript%3Ealert(1)&id=1 HTTP/1.1

It can be deduced from above that during the live capture, I logged into dvwa via **/dvwa/login** endpoint. Afterwards, I went to the XSS section of the available web vulnerabilities. Following the TCP and HTTP streams produced different results. The former's response body is in a non readable format while the latter is clearly human readable.

<pre> HTTP/1.1 200 OK Date: Fri, 07 Nov 2025 20:08:57 GMT Server: Apache/2.2.14 (Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL/0.9.8k Phusion_Passenger/4.0.38 mod_perl/2.0.4 Perl/v5.10.1 Last-Modified: Fri, 31 Jul 2015 02:55:52 GMT ETag: "45f13-6da3-51c22f5365e00" Accept-Ranges: bytes Vary: Accept-Encoding Content-Encoding: gzip Content-Length: 4785 Keep-Alive: timeout=15, max=100 Connection: Keep-Alive Content-Type: text/html </pre>	<pre> <!DOCTYPE HTML> <html> <head> <title>owaspbwa OWASP Broken Web Applications</title> <link rel="stylesheet" href="index.css" type="text/css" media="screen" /> <script type="text/javascript" src="/jquery.min.js"></script> <!--http://ajax.googleapis.com/ajax/libs/jquery/1.3.2/jquery.min.js pulled april 15 2011--> <script type="text/javascript" src="animatedcollapse.js"> * Animated Collapsible DIV v2.4- (c) Dynamic Drive DHTML code library (www.dynamicdrive.com) This notice MUST stay intact for legal use * Visit Dynamic Drive at http://www.dynamicdrive.com/ for this script and 100s more </script> <script type="text/javascript"> animatedcollapse.addDiv('webgoat', 'fade=1') animatedcollapse.addDiv('webgoat_net', 'fade=1') animatedcollapse.addDiv('swingset', 'fade=1') animatedcollapse.addDiv('swingset_interactive', 'fade=1') animatedcollapse.addDiv('mutillidae', 'fade=1') animatedcollapse.addDiv('dvwa', 'fade=1') animatedcollapse.addDiv('ghost', 'fade=1') animatedcollapse.addDiv('bwapp', 'fade=1') </pre>
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One of the packets was the request I made while attempting to test a XSS vulnerability. The payload **<script>alert(1)</script>** is encoded in Wireshark. The response body somewhat tells us that the attack was successful.

<pre> </html>GET /dvwa/vulnerabilities/xss_r/?name=%3Cscript%3Ealert(1)&id=1 HTTP/1.1 Host: 192.168.1.149 User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8 Accept-Language: en-US,en;q=0.5 Accept-Encoding: gzip, deflate Connection: keep-alive Referer: http://192.168.1.149/dvwa/vulnerabilities/xss_r/ Cookie: security=low; PHPSESSID=c0n4npahemsv88pege1k70; acopendivids=swingset,jotto,pqbb2,redmine; acgroupswithpersist=nada Upgrade-Insecure-Requests: 1 DNT: 1 Sec-GPC: 1 Priority: u=0, i </pre>	<pre> HTTP/1.1 200 OK Date: Fri, 07 Nov 2025 20:09:36 GMT Server: Apache/2.2.14 (Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL/0.9.8k Phusion_Passenger/4.0.38 mod_perl/2.0.4 Perl/v5.10.1 X-Powered-By: PHP/5.3.2-1ubuntu4.30 Expires: Tue, 23 Jun 2009 12:00:00 GMT </pre>
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I tried some other filters to display **DSN** packets and keywords within the packets.

dns							
No.	Time	Source	Destination	Protocol	Length	Info	
36	2025-11-07 21:08:53.062762787	192.168.1.144	192.168.1.1	DNS	85	Standard query 0x5778 A push.services.mozilla.com	
37	2025-11-07 21:08:53.063026021	192.168.1.144	192.168.1.1	DNS	87	Standard query 0x11fd A safebrowsing.googleapis.com	
38	2025-11-07 21:08:53.063118144	192.168.1.144	192.168.1.1	DNS	85	Standard query 0xf07e AAAA push.services.mozilla.com	
39	2025-11-07 21:08:53.063724227	192.168.1.144	192.168.1.1	DNS	87	Standard query 0x80fc AAAA safebrowsing.googleapis.com	
41	2025-11-07 21:08:53.358210072	192.168.1.144	192.168.1.1	DNS	88	Standard query 0x22fa A contile.services.mozilla.com	
42	2025-11-07 21:08:53.358506675	192.168.1.144	192.168.1.1	DNS	88	Standard query 0x9df8 AAAA contile.services.mozilla.com	
43	2025-11-07 21:08:53.359622204	192.168.1.144	192.168.1.1	DNS	95	Standard query 0x2604 A content-signature-2.cdn.mozilla.net	
44	2025-11-07 21:08:53.359739450	192.168.1.144	192.168.1.1	DNS	95	Standard query 0xfb09 AAAA content-signature-2.cdn.mozilla.net	
45	2025-11-07 21:08:53.443093184	192.168.1.144	192.168.1.1	DNS	97	Standard query 0xc582 A firefox.settings.services.mozilla.com	
46	2025-11-07 21:08:53.443311632	192.168.1.144	192.168.1.1	DNS	97	Standard query 0x3080 AAAA firefox.settings.services.mozilla.com	
47	2025-11-07 21:08:53.443956453	192.168.1.144	192.168.1.1	DNS	88	Standard query 0xe3d8 A profile.accounts.firefox.com	
48	2025-11-07 21:08:53.444052212	192.168.1.144	192.168.1.1	DNS	88	Standard query 0x5cda AAAA profile.accounts.firefox.com	
49	2025-11-07 21:08:53.712190532	192.168.1.1	192.168.1.144	DNS	104	Standard query response 0x22fa A contile.services.mozilla.com	
50	2025-11-07 21:08:53.754931334	192.168.1.144	192.168.1.1	DNS	86	Standard query 0xf425 A token.services.mozilla.com	
51	2025-11-07 21:08:53.755113112	192.168.1.144	192.168.1.1	DNS	86	Standard query 0xb127 AAAA token.services.mozilla.com	
52	2025-11-07 21:08:53.972939827	192.168.1.1	192.168.1.144	DNS	197	Standard query response 0xc582 A firefox.settings.services.mozilla.com	
53	2025-11-07 21:08:53.974743934	192.168.1.1	192.168.1.144	DNS	189	Standard query response 0xe3d8 A profile.accounts.firefox.com	

http contains "dvwa"							
No.	Time	Source	Destination	Protocol	Length	Info	
400	2025-11-07 21:09:01.967473380	192.168.1.144	192.168.1.149	HTTP	455	GET /dvwa HTTP/1.1	
401	2025-11-07 21:09:01.970578675	192.168.1.149	192.168.1.144	HTTP	757	HTTP/1.1 301 Moved Permanently (text/html)	
403	2025-11-07 21:09:02.018747593	192.168.1.144	192.168.1.149	HTTP	456	GET /dvwa/ HTTP/1.1	
406	2025-11-07 21:09:02.069513353	192.168.1.144	192.168.1.149	HTTP	525	GET /dvwa/login.php HTTP/1.1	
424	2025-11-07 21:09:02.319321363	192.168.1.144	192.168.1.149	HTTP	540	GET /dvwa/dvwa/css/login.css HTTP/1.1	
437	2025-11-07 21:09:02.323357740	192.168.1.144	192.168.1.149	HTTP	632	GET /dvwa/dvwa/images/login_logo.png HTTP/1.1	
1125	2025-11-07 21:09:16.528853654	192.168.1.144	192.168.1.149	HTTP	750	POST /dvwa/login.php HTTP/1.1 (application/x-www-form-urlencoded)	
1134	2025-11-07 21:09:16.620475257	192.168.1.144	192.168.1.149	HTTP	609	GET /dvwa/index.php HTTP/1.1	
1150	2025-11-07 21:09:16.813060459	192.168.1.144	192.168.1.149	HTTP	539	GET /dvwa/dvwa/css/main.css HTTP/1.1	
1156	2025-11-07 21:09:16.817813363	192.168.1.144	192.168.1.149	HTTP	526	GET /dvwa/dvwa/js/dvwaPage.js HTTP/1.1	
1160	2025-11-07 21:09:16.834426928	192.168.1.144	192.168.1.149	HTTP	596	GET /dvwa/dvwa/images/logo.png HTTP/1.1	
1167	2025-11-07 21:09:16.869366646	192.168.1.144	192.168.1.149	HTTP	460	GET /dvwa/dvwa/js/dvwaPage.js HTTP/1.1	
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1403	2025-11-07 21:09:24.177459150	192.168.1.144	192.168.1.149	HTTP	622	GET /dvwa/vulnerabilities/xss_r/?name=%3Cscript%3Ealert%28%22document.cookie%22%29 HTTP/1.1	
1649	2025-11-07 21:09:34.623965209	192.168.1.144	192.168.1.149	HTTP	680	GET /dvwa/vulnerabilities/xss_r/?name=%3Cscript%3Ealert%28%22document.cookie%22%29 HTTP/1.1	

OBSERVATIONS

The following are some of my observations during the analysis of the captured packets:

- Quite a lot of packets were captured in so little time.
- I observed that packets were immediately being captured after I clicked on the capture button in Wireshark even though the only apps running were Wireshark and Terminal. My guess is that there are some services that are communicating over the internet in the background and I plan on making further research on this.
- TCP response was unreadable unlike HTTP. It does not seem encryption related and I will also make findings on this.

In conclusion, I reaffirmed that analyzing packets is a really powerful method of investigating what has taken place on a computer, incidents or breaches and in general, gaining an insight into communications that have taken place over the internet.